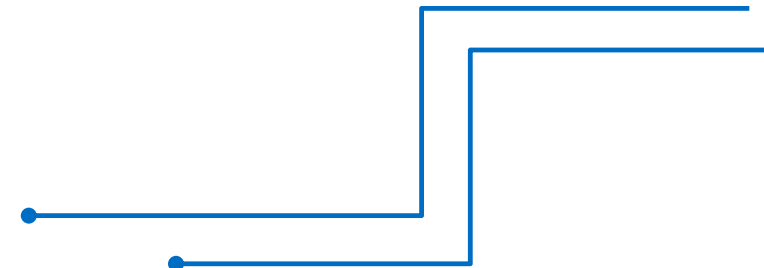


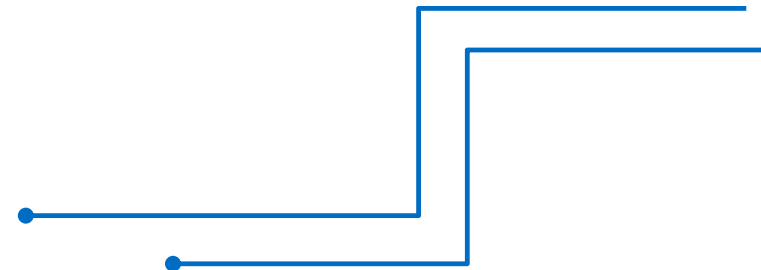
Courses

- ❖ **What is Matplotlib?**
- ❖ **Installation of Matplotlib**
- ❖ **Import Matplotlib**
- ❖ **Checking Matplotlib Version**
- ❖ **Essential Role of Pyplot in Matplotlib**
- ❖ **Types of Plots**



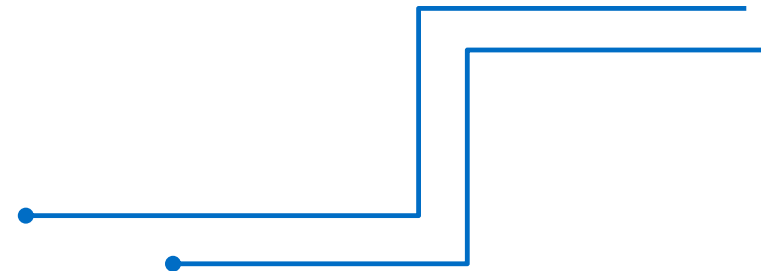
What is Matplot

- ❖ Matplotlib is a low level graph plotting library in python that serves as a visualization utility.
- ❖ Matplotlib was created by John D. Hunter.
- ❖ Matplotlib is open source and we can use it freely.
- ❖ Matplotlib is mostly written in python, a few segments are written in C, Objective-C and Javascript for Platform compatibility.



Installation of Matplotlib

```
C:\Users\Your Name>pip install matplotlib
```

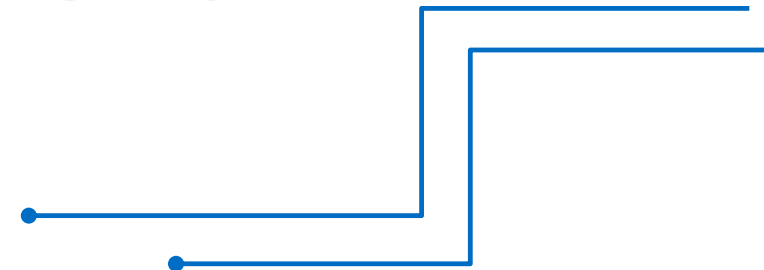
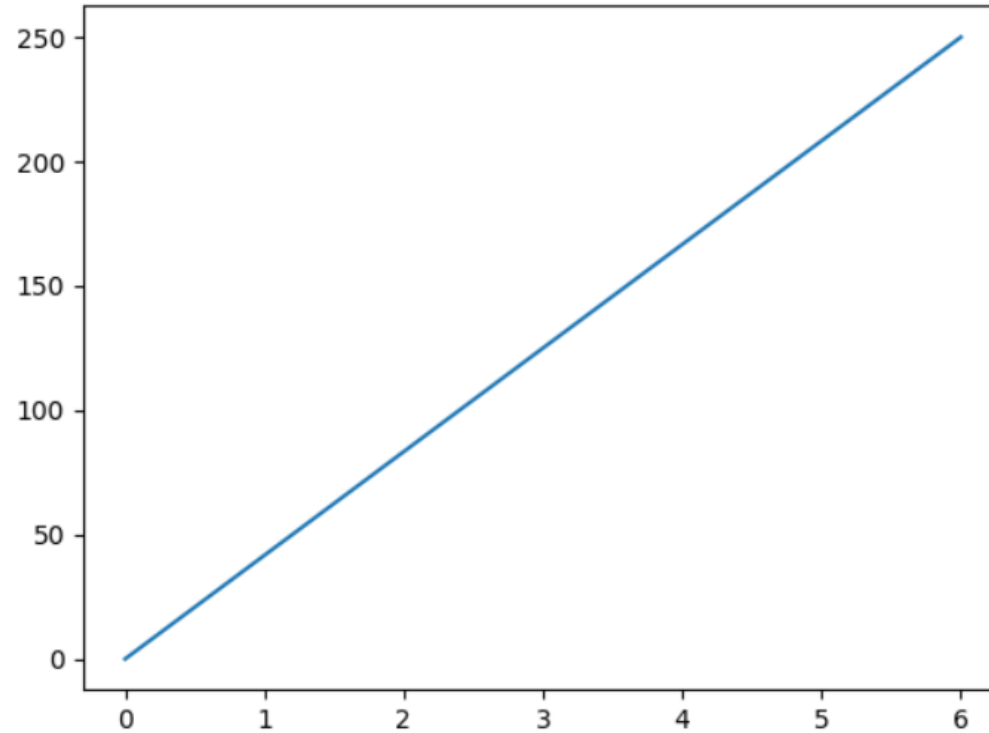


Line Chart

```
import matplotlib.pyplot as plt
import numpy as np

xpoints = np.array([0, 6])
ypoints = np.array([0, 250])

plt.plot(xpoints, ypoints)
plt.show()
```

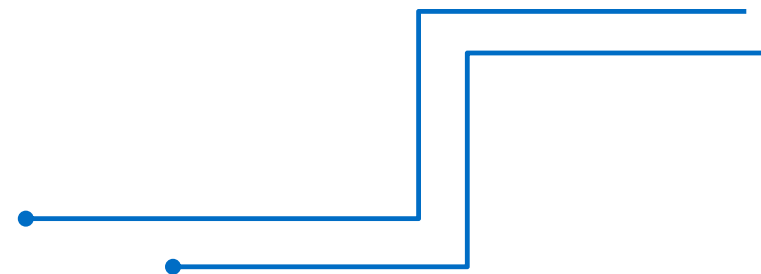
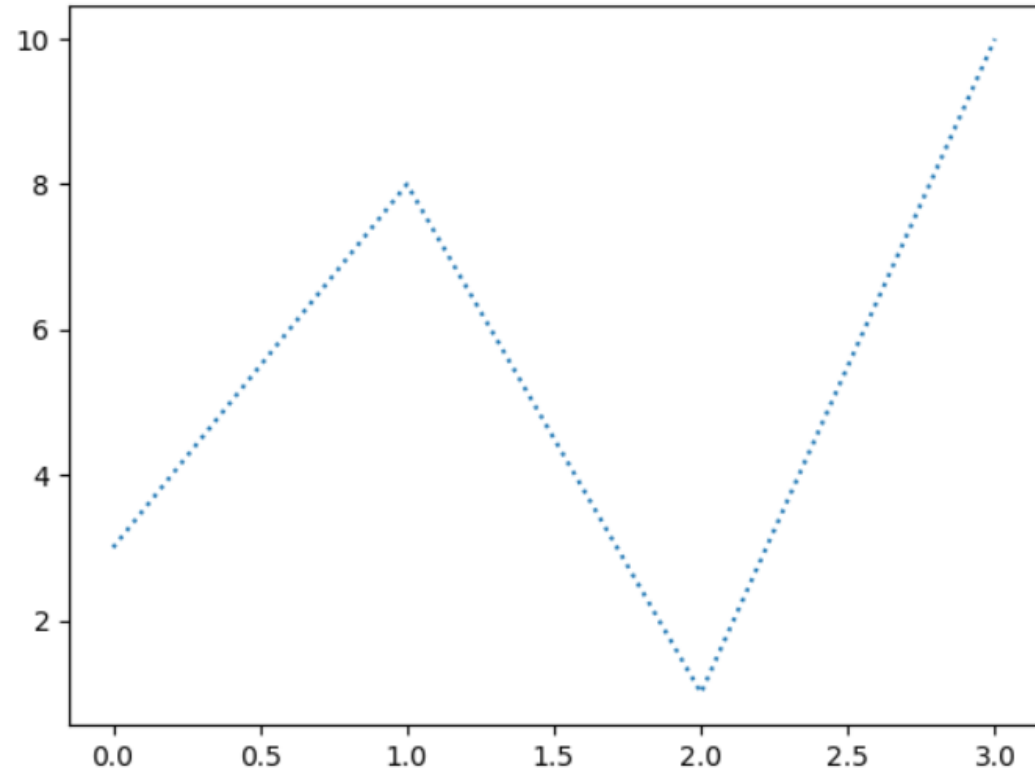


Line Chart

```
import matplotlib.pyplot as plt
import numpy as np

ypoints = np.array([3, 8, 1, 10])

plt.plot(ypoints, linestyle = 'dotted')
plt.show()
```

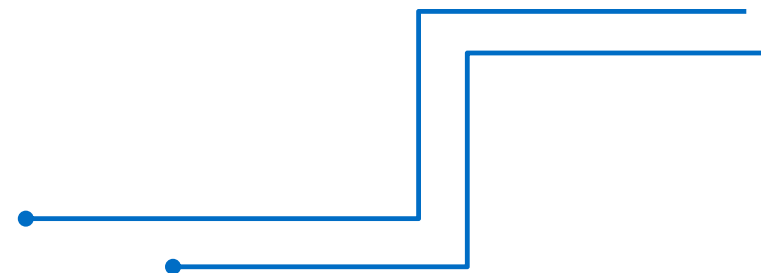
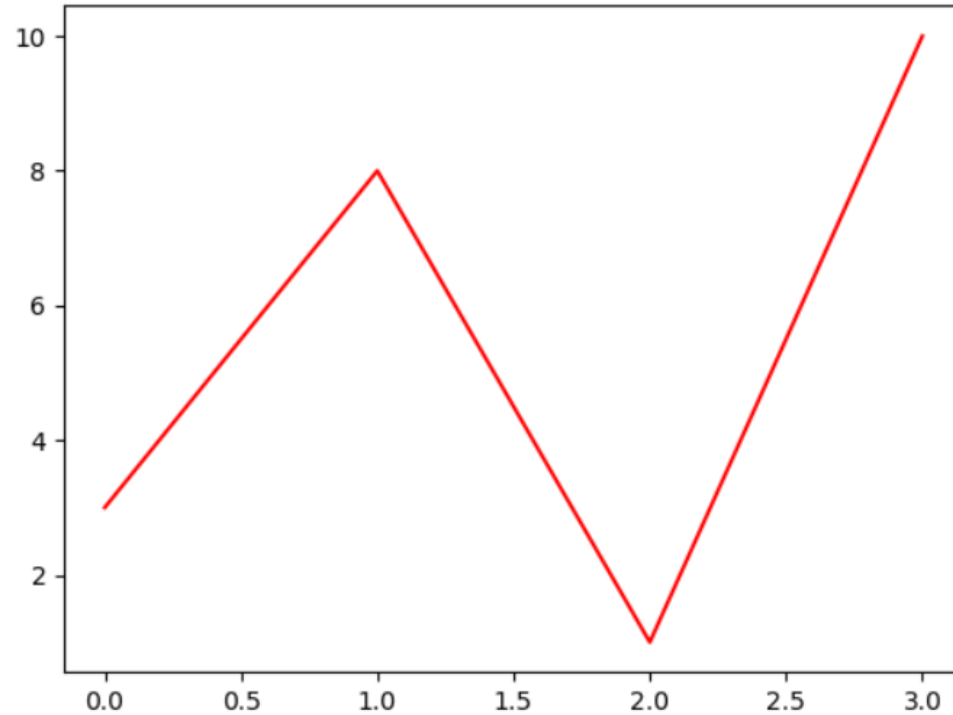


Line Chart

```
import matplotlib.pyplot as plt
import numpy as np

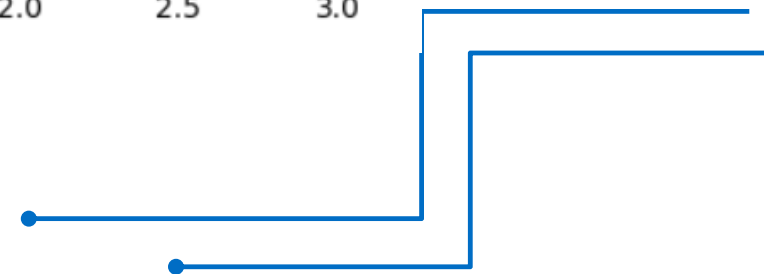
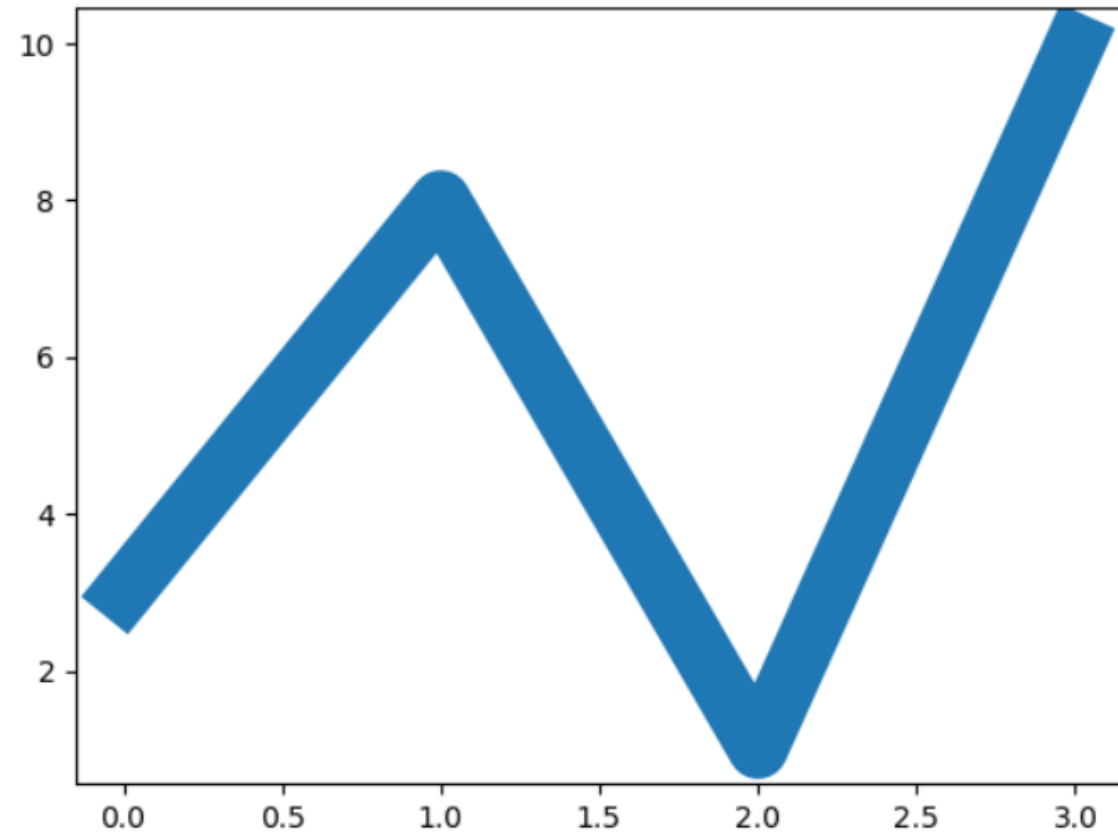
ypoints = np.array([3, 8, 1, 10])

plt.plot(ypoints, color = 'r')
plt.show()
```



Line Chart

```
import matplotlib.pyplot as plt  
import numpy as np  
  
ypoints = np.array([3, 8, 1, 10])  
  
plt.plot(ypoints, linewidth = '20.5')  
plt.show()
```



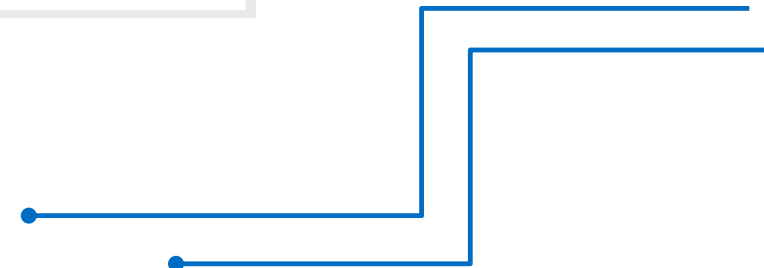
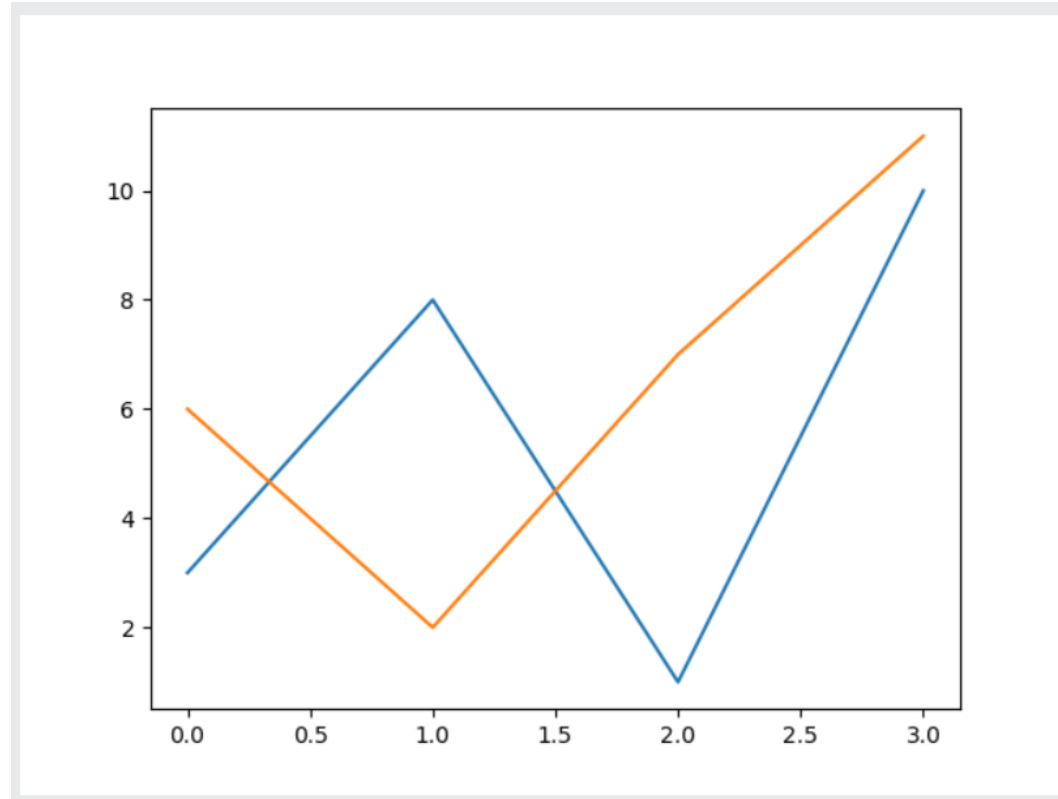
Multi Line

```
import matplotlib.pyplot as plt  
import numpy as np
```

```
y1 = np.array([3, 8, 1, 10])  
y2 = np.array([6, 2, 7, 11])
```

```
plt.plot(y1)  
plt.plot(y2)
```

```
plt.show()
```



Matplotlib Labels and Title

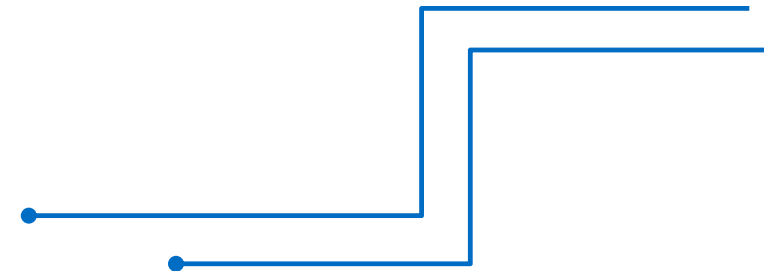
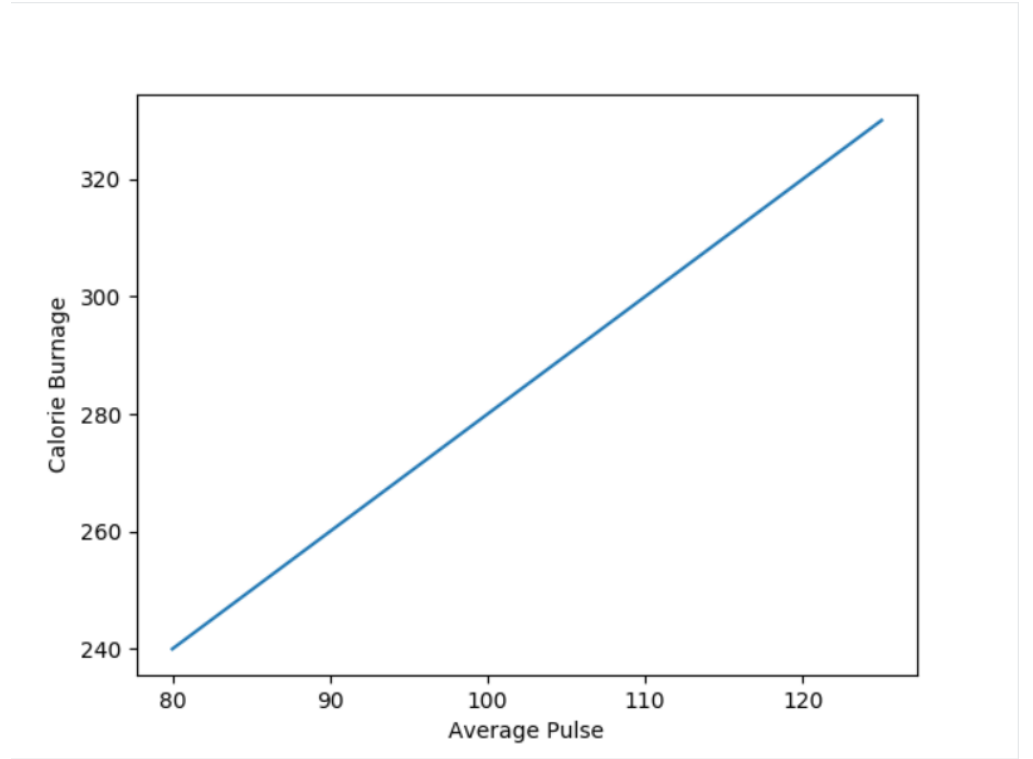
```
import numpy as np
import matplotlib.pyplot as plt

x = np.array([80, 85, 90, 95, 100, 105, 110, 115, 120, 125])
y = np.array([240, 250, 260, 270, 280, 290, 300, 310, 320, 330])

plt.plot(x, y)

plt.xlabel("Average Pulse")
plt.ylabel("Calorie Burnage")

plt.show()
```



Create a Title for a Plot

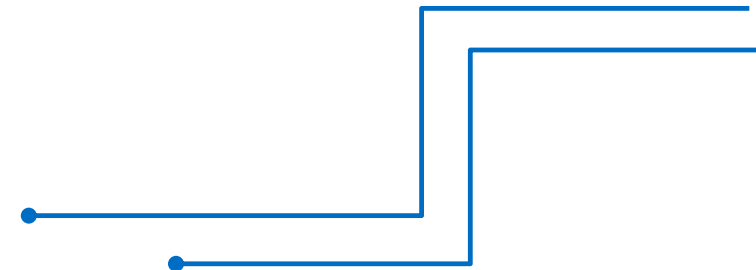
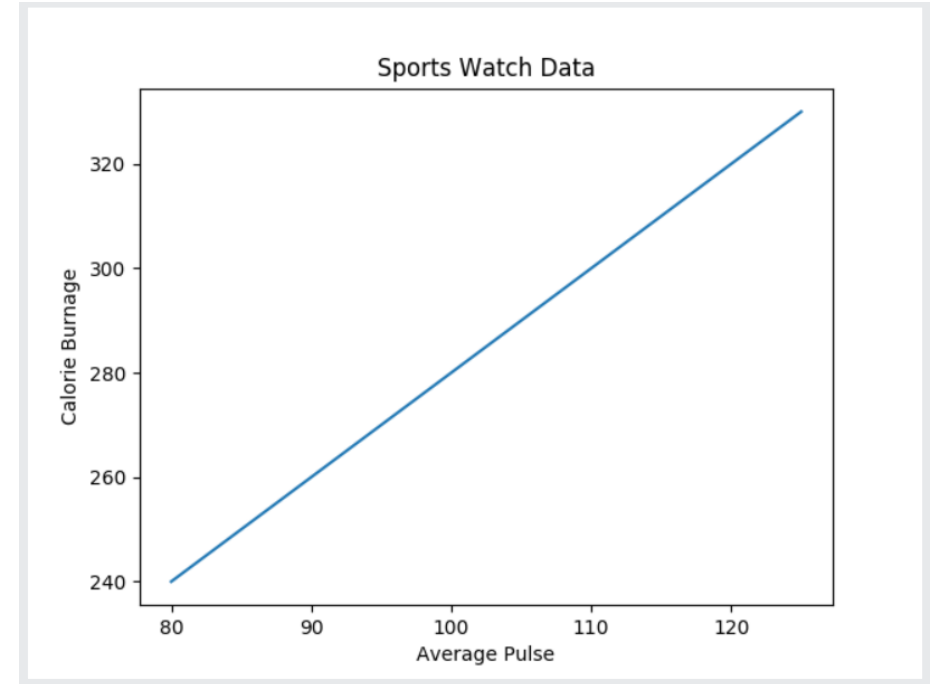
```
import numpy as np
import matplotlib.pyplot as plt

x = np.array([80, 85, 90, 95, 100, 105, 110, 115, 120, 125])
y = np.array([240, 250, 260, 270, 280, 290, 300, 310, 320, 330])

plt.plot(x, y)

plt.title("Sports Watch Data")
plt.xlabel("Average Pulse")
plt.ylabel("Calorie Burnage")

plt.show()
```

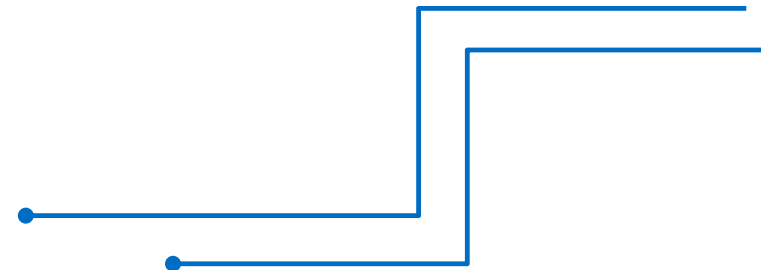
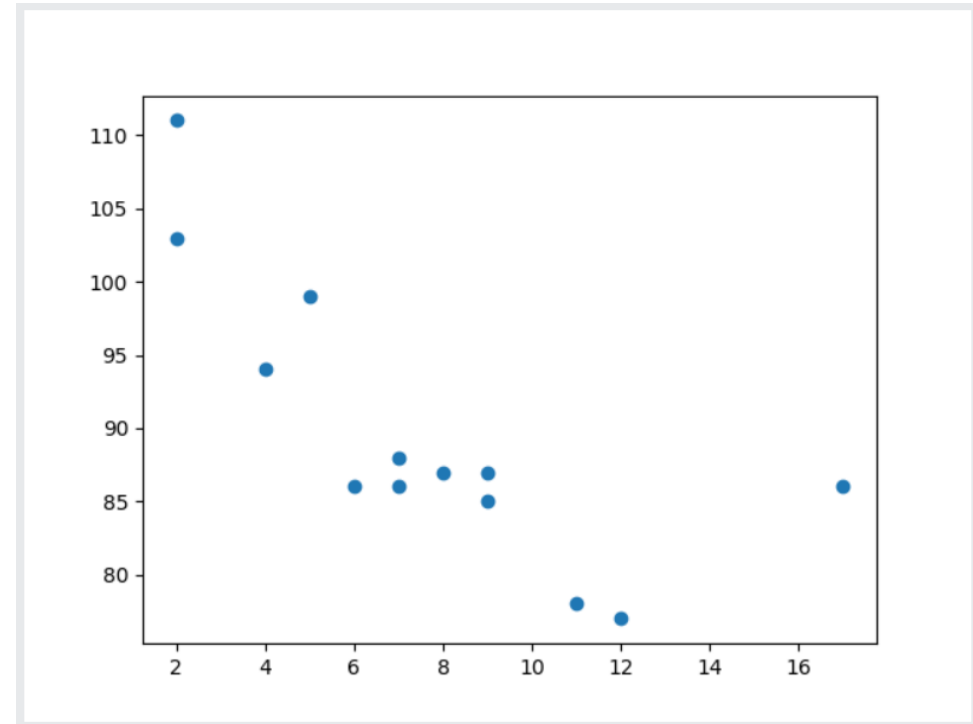


Creating Scatter Plots

```
import matplotlib.pyplot as plt
import numpy as np

x = np.array([5,7,8,7,2,17,2,9,4,11,12,9,6])
y = np.array([99,86,87,88,111,86,103,87,94,78,77,85,86])

plt.scatter(x, y)
plt.show()
```



Use the ColorMap

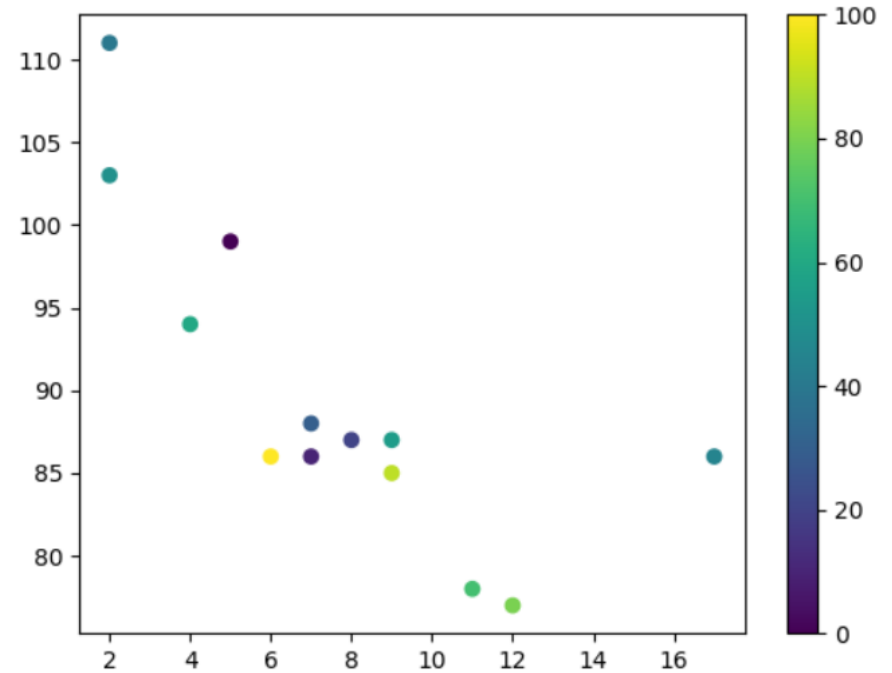
```
import matplotlib.pyplot as plt
import numpy as np

x = np.array([5,7,8,7,2,17,2,9,4,11,12,9,6])
y = np.array([99,86,87,88,111,86,103,87,94,78,77,85,86])
colors =
np.array([0, 10, 20, 30, 40, 45, 50, 55, 60, 70, 80, 90, 100])

plt.scatter(x, y, c=colors, cmap='viridis')

plt.colorbar()

plt.show()
```



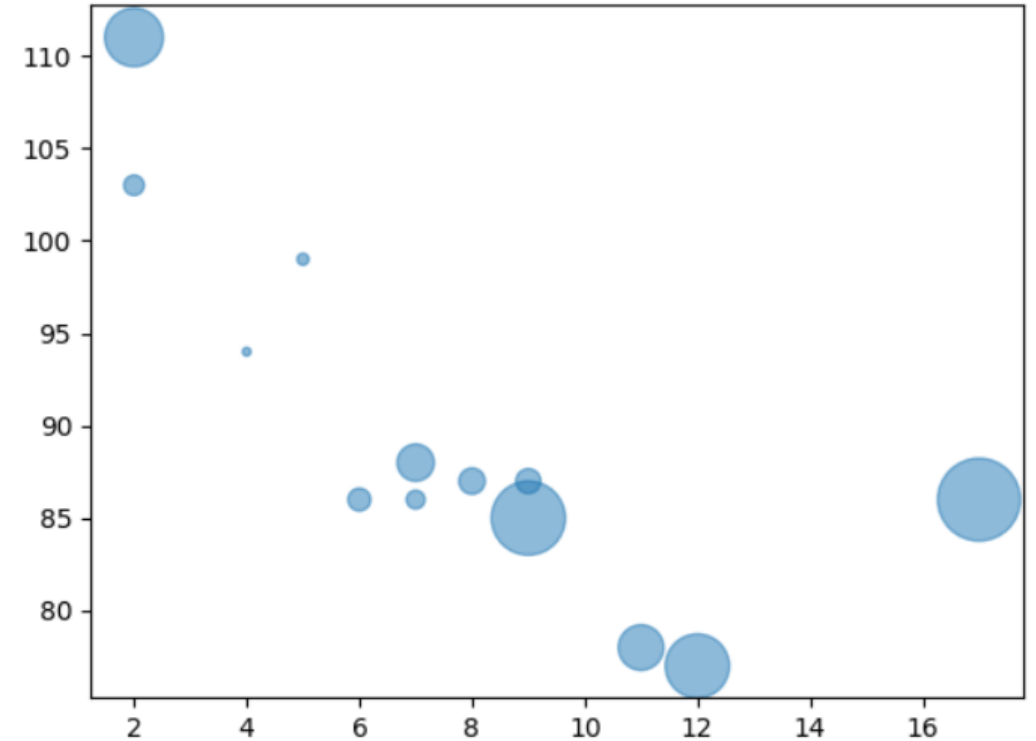
Use the Alpha

```
import matplotlib.pyplot as plt
import numpy as np

x = np.array([5,7,8,7,2,17,2,9,4,11,12,9,6])
y = np.array([99,86,87,88,111,86,103,87,94,78,77,85,86])
sizes
= np.array([20,50,100,200,500,1000,60,90,10,300,600,800,75])

plt.scatter(x, y, s=sizes, alpha=0.5)

plt.show()
```



Use the Alpha

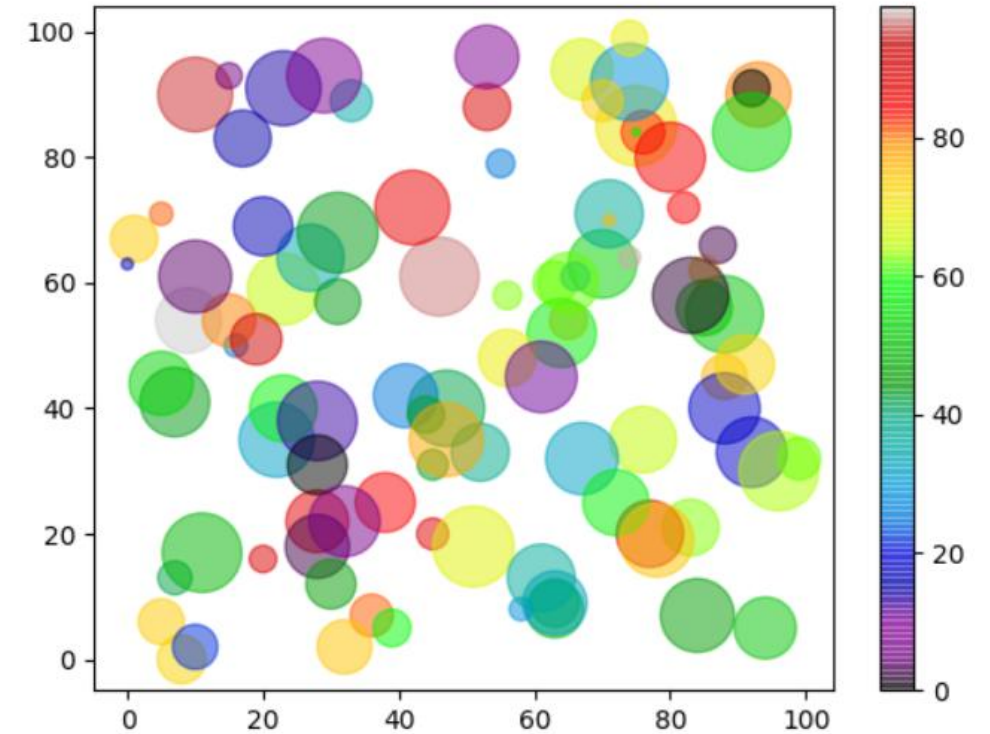
```
import matplotlib.pyplot as plt
import numpy as np

x = np.random.randint(100, size=(100))
y = np.random.randint(100, size=(100))
colors = np.random.randint(100, size=(100))
sizes = 10 * np.random.randint(100, size=(100))

plt.scatter(x, y, c=colors, s=sizes, alpha=0.5,
            cmap='nipy_spectral')

plt.colorbar()

plt.show()
```

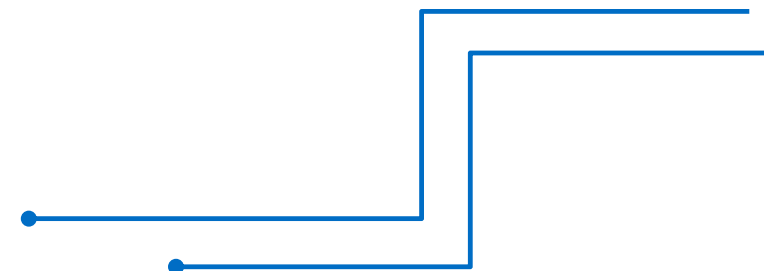
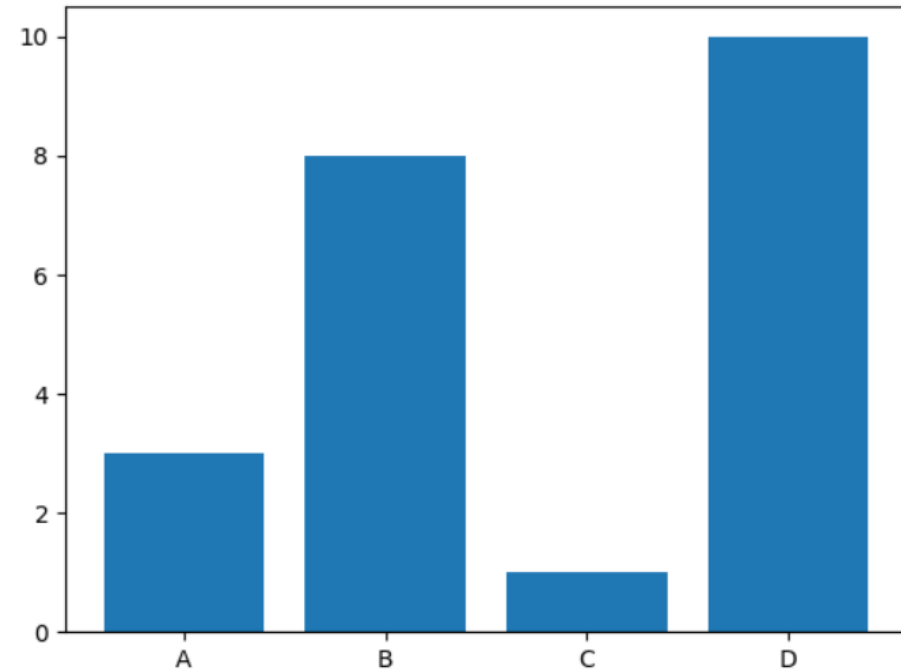


Creating Bars

```
import matplotlib.pyplot as plt  
import numpy as np
```

```
x = np.array(["A", "B", "C", "D"])  
y = np.array([3, 8, 1, 10])
```

```
plt.bar(x,y)  
plt.show()
```

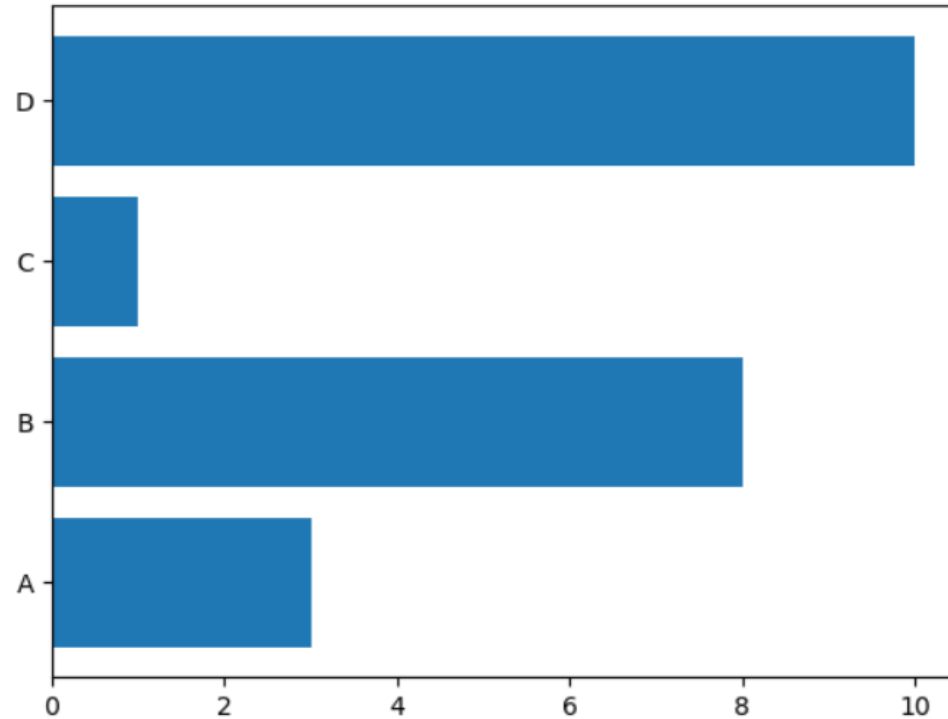


Horizontal Bars

```
import matplotlib.pyplot as plt  
import numpy as np
```

```
x = np.array(["A", "B", "C", "D"])  
y = np.array([3, 8, 1, 10])
```

```
plt.barh(x, y)  
plt.show()
```

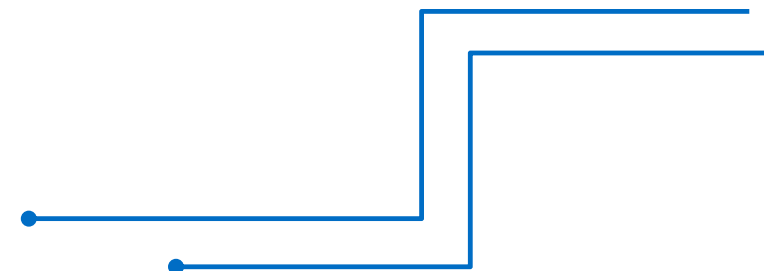
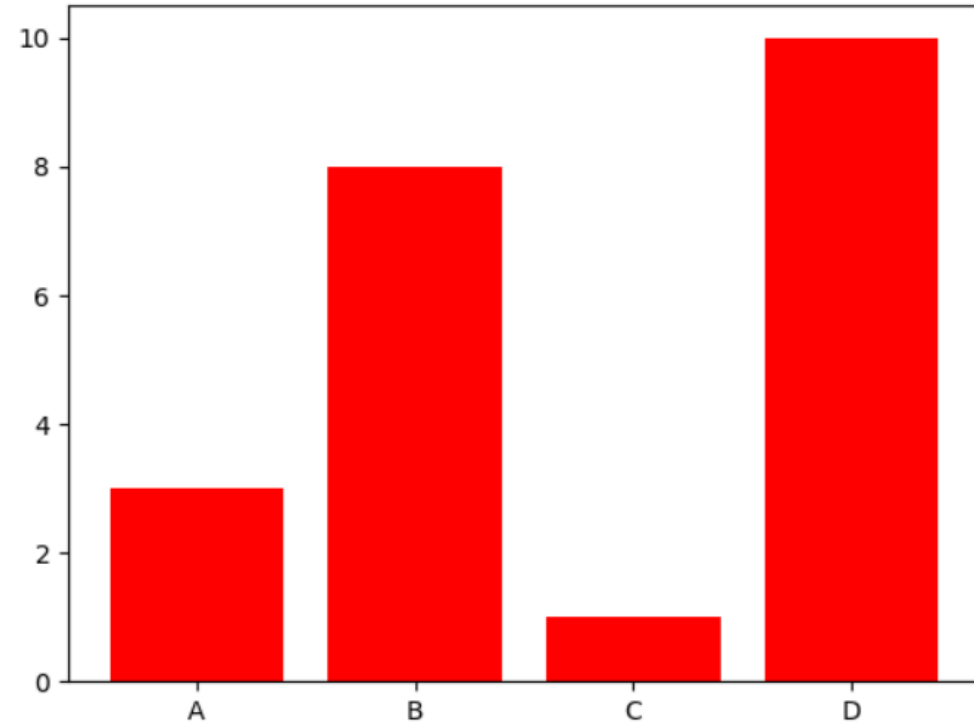


Bar Color

```
import matplotlib.pyplot as plt  
import numpy as np
```

```
x = np.array(["A", "B", "C", "D"])  
y = np.array([3, 8, 1, 10])
```

```
plt.bar(x, y, color = "red")  
plt.show()
```

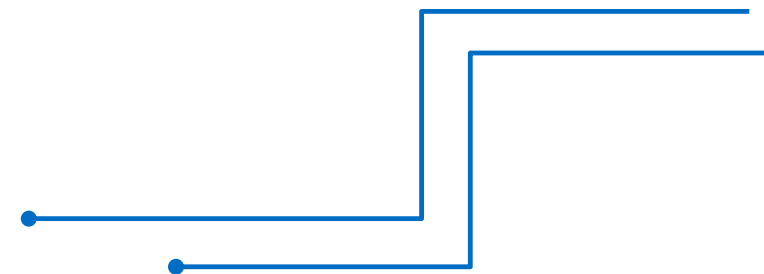
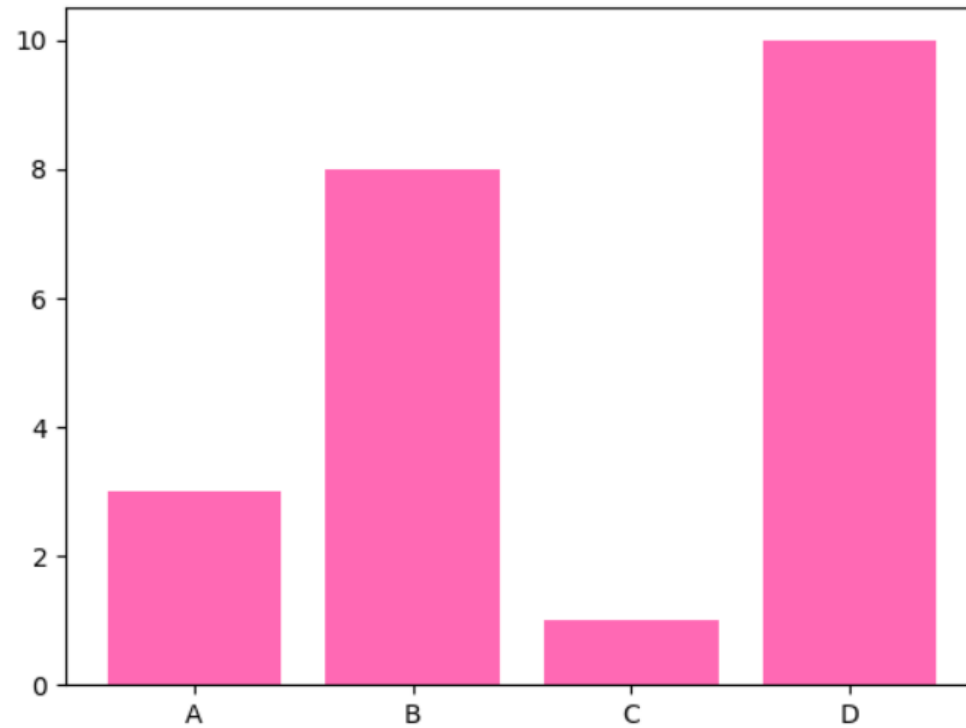


Bar Color

```
import matplotlib.pyplot as plt  
import numpy as np
```

```
x = np.array(["A", "B", "C", "D"])  
y = np.array([3, 8, 1, 10])
```

```
plt.bar(x, y, color = "hotpink")  
plt.show()
```

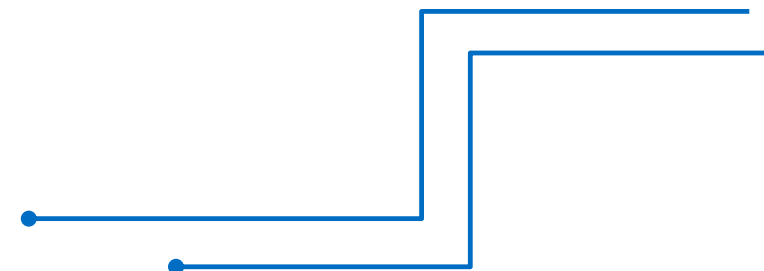
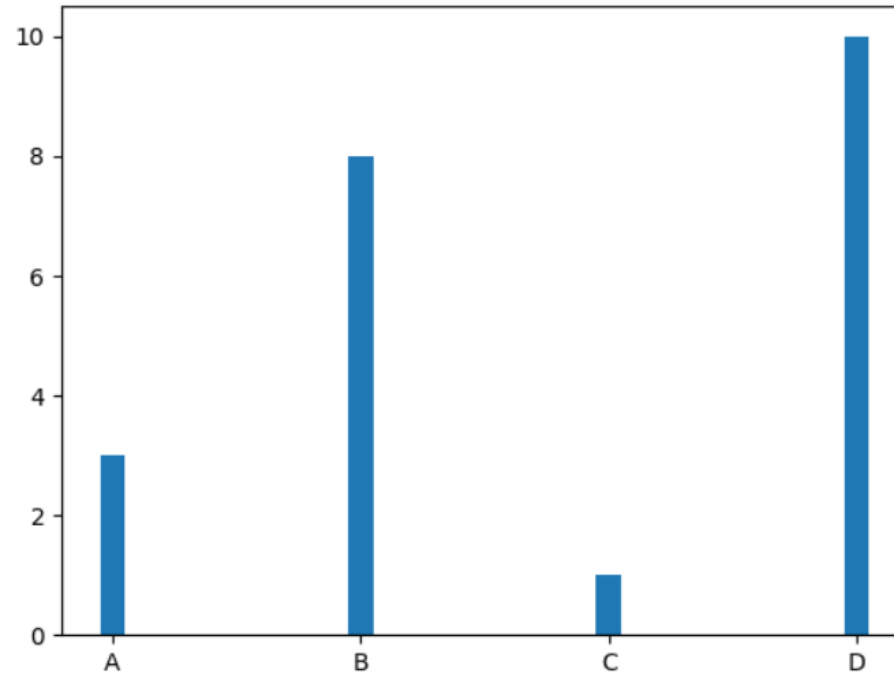


Width

```
import matplotlib.pyplot as plt  
import numpy as np
```

```
x = np.array(["A", "B", "C", "D"])  
y = np.array([3, 8, 1, 10])
```

```
plt.bar(x, y, width = 0.1)  
plt.show()
```

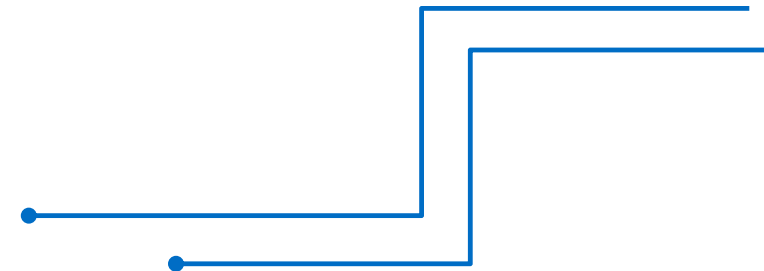
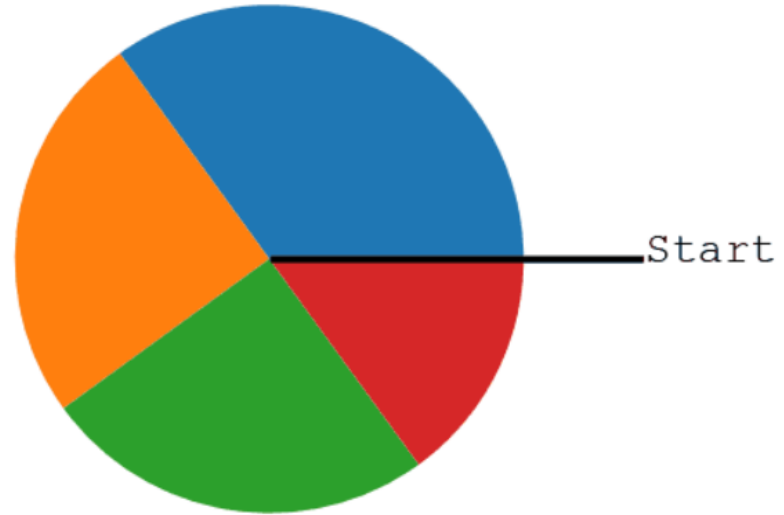


Creating Pie Charts

```
import matplotlib.pyplot as plt  
import numpy as np
```

```
y = np.array([35, 25, 25, 15])
```

```
plt.pie(y)  
plt.show()
```

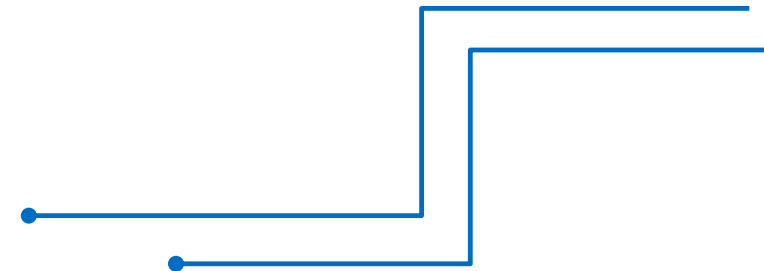
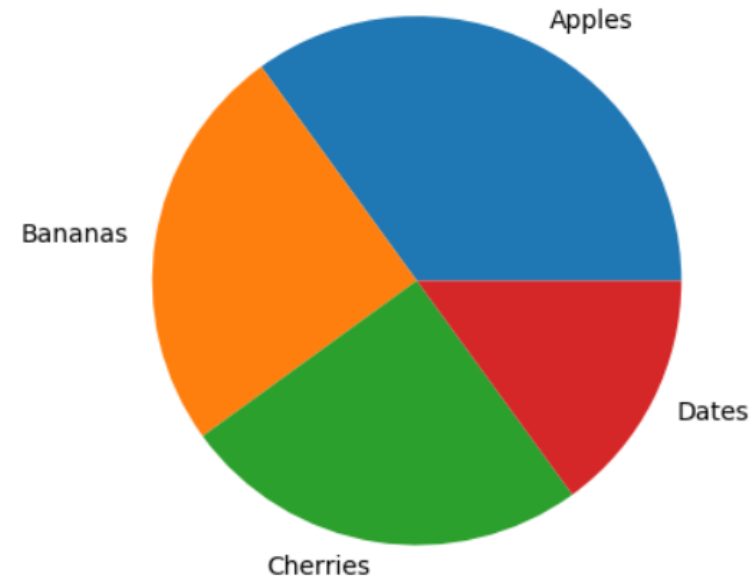


Creating Pie Charts Labels

```
import matplotlib.pyplot as plt
import numpy as np

y = np.array([35, 25, 25, 15])
mylabels =
["Apples", "Bananas", "Cherries", "Dates"]

plt.pie(y, labels = mylabels)
plt.show()
```

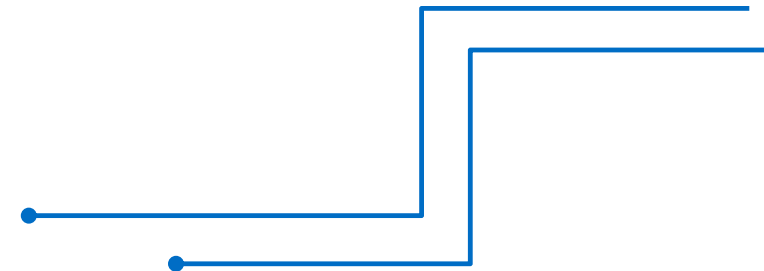
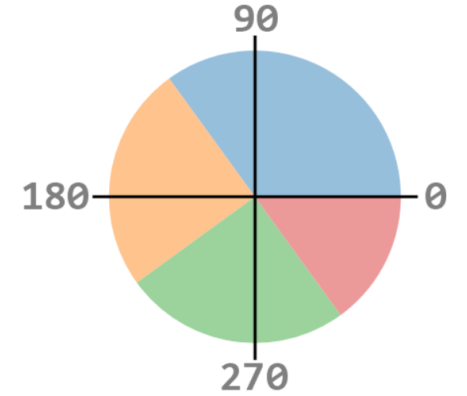
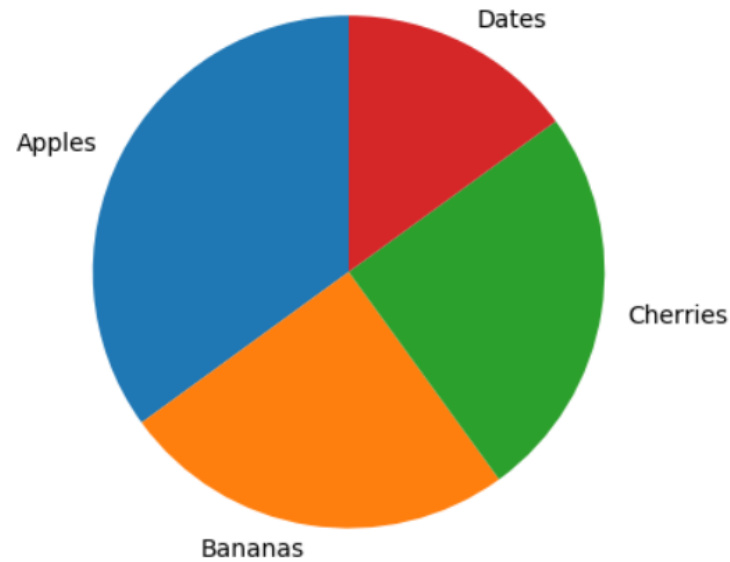


Start Angle

```
import matplotlib.pyplot as plt
import numpy as np

y = np.array([35, 25, 25, 15])
mylabels =
["Apples", "Bananas", "Cherries", "Dates"]

plt.pie(y, labels = mylabels, startangle = 90)
plt.show()
```

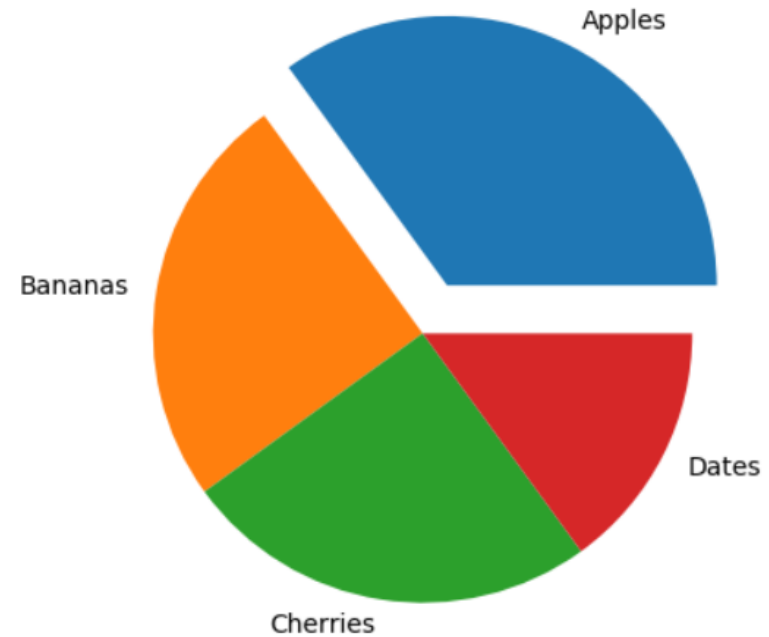


Explode

```
import matplotlib.pyplot as plt
import numpy as np

y = np.array([35, 25, 25, 15])
mylabels =
["Apples", "Bananas", "Cherries", "Dates"]
myexplode = [0.2, 0, 0, 0]

plt.pie(y, labels = mylabels, explode = myexplode)
plt.show()
```

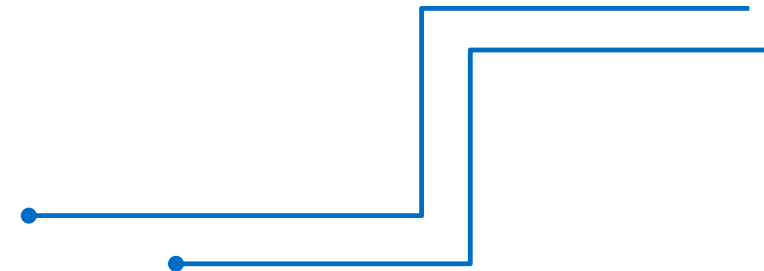
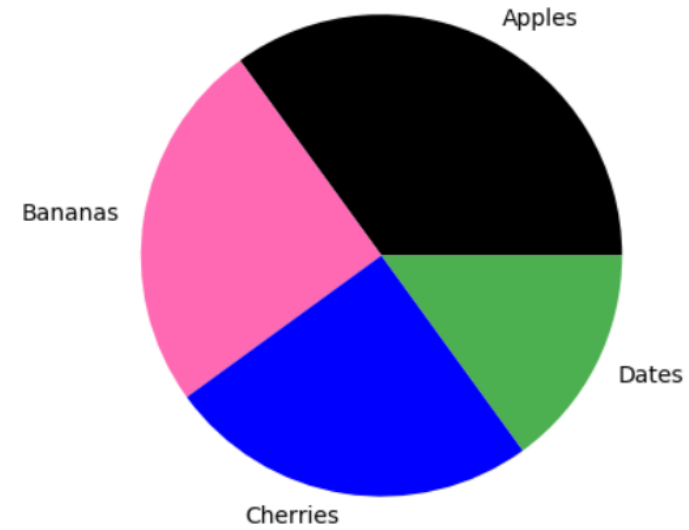


Colors

```
import matplotlib.pyplot as plt
import numpy as np

y = np.array([35, 25, 25, 15])
mylabels =
["Apples", "Bananas", "Cherries", "Dates"]
mycolors = ["black", "hotpink", "b", "#4CAF50"]

plt.pie(y, labels = mylabels, colors = mycolors)
plt.show()
```



Legend

```
import matplotlib.pyplot as plt
import numpy as np

y = np.array([35, 25, 25, 15])
mylabels =
["Apples", "Bananas", "Cherries", "Dates"]

plt.pie(y, labels = mylabels)
plt.legend()
plt.show()
```

